HR ANALYTICS

PROJECT REPORT

Turning Data into Insights and Insights into Impact

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Dataset
IBM HR Analytics
1,470 Employees | 35 Variables

4	20+	30+
Dashboards	DAX Measures	Visualizations

Executive Summary

Employee attrition remains one of the most significant challenges facing organizations. High turnover increases recruitment costs, affects productivity, reduces employee morale, and disrupts business operations.

The objective of this project was to analyze employee data to identify patterns associated with workforce performance and employee attrition. Rather than focusing only on descriptive metrics, the analysis aimed to answer business-critical questions that support informed decision-making.

The project resulted in four interactive dashboards that provide executives and HR professionals with a comprehensive view of workforce composition, employee retention, satisfaction, compensation, and workforce planning.

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1. Introduction

Human Resources departments generate large volumes of employee data every day. When analyzed effectively, this data provides valuable insights into workforce trends, employee engagement, and organizational performance.

This project demonstrates how HR data can be transformed into actionable insights that enable organizations to make informed workforce decisions.

2. Business Problem

Employee attrition has both direct and indirect costs. Organizations often struggle to answer critical workforce questions without data-driven tools.

> **Without answers to critical workforce questions, organizations risk making reactive rather than proactive decisions; costing time, resources, and talent.**

Key unanswered questions include:

- Which employees are most likely to leave?
- Which departments experience the highest turnover?
- What factors contribute to employee attrition?
- Are compensation and career progression influencing employee retention?
- Which workforce groups require immediate attention?

3. Project Objectives

The project aimed to:

- Understand workforce composition and demographics
- Measure and analyze employee attrition across multiple dimensions
- Identify factors associated with employee turnover
- Evaluate employee satisfaction and compensation levels
- Support data-driven workforce planning
- Provide actionable recommendations based on real data

4. Dataset Overview

The analysis was performed using the IBM HR Analytics Employee Attrition dataset, a widely recognized benchmark dataset for HR analytics.

Property	Detail
Total Records	1,470 employees
Total Variables	35 features
Target Variable	Attrition (Yes/No)
Source	IBM HR Analytics Dataset
Format	CSV
Tool Used	Power BI Desktop

Key Variables

The dataset includes:

- Employee Demographics: Age, Gender, Marital Status, Education
- Employment Details: Department, Job Role, Job Level, Business Travel
- Compensation: Monthly Income, Salary Hike, Stock Options
- Performance: Performance Rating, Training Times
- Satisfaction: Job Satisfaction, Environment Satisfaction, Work-Life Balance, Relationship Satisfaction
- Tenure: Years at Company, Years in Current Role, Years Since Last Promotion, Years with Manager
- Target: Attrition (Yes/No)

5. Data Preparation

The dataset was reviewed to ensure data quality before analysis. The preparation process included:

- Verifying and correcting data types across all columns
- Checking for and confirming absence of missing values
- Reviewing for duplicate records
- Creating calculated columns for grouped categories (Age Group, Income Band, Distance Band, Career Stage, Tenure Band)
- Developing DAX measures for key performance indicators
- Standardizing fields used for reporting and cross-filtering

> **These preparation steps improved data reliability and ensured consistency and accuracy across all four dashboards.**

6. Data Modeling

A dedicated measures table (~Measure) was created to organize all DAX calculations separately from the raw data table. This approach improves model performance and maintainability.

Core DAX Measures

Measure	Purpose
Total Employees	Count of all employee records
Active Employees	Count of employees who have not left
Employees Left	Count of employees who left
Attrition Rate	Percentage of employees who left
Avg Monthly Income	Average employee monthly income
Avg Job Satisfaction	Average satisfaction score (out of 5)
Avg Years at Company	Average tenure across all employees
Group Attrition Rate	Attrition rate within any filtered group
High Satisfaction Attrition	Attrition rate for highly satisfied employees
Recent Promotion Attrition	Attrition rate for recently promoted employees

7. Dashboard Design

The report consists of four interactive dashboards built in Power BI Desktop. Each dashboard includes cross-filtering slicers, data-driven insights, and actionable recommendations.

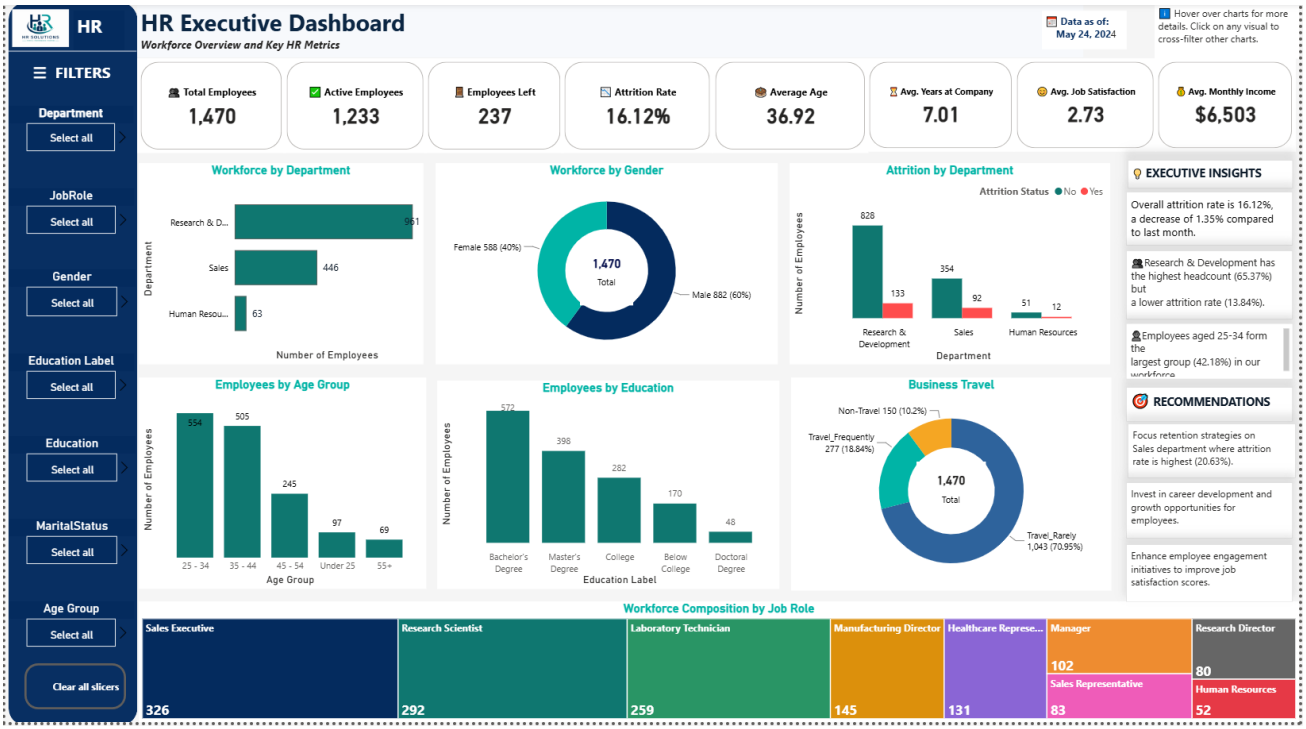
Dashboard 1 — HR Executive Overview

Purpose: Provide executives with a high-level snapshot of overall workforce performance.

Key Metrics:

- Total Employees: 1,470 | Active Employees: 1,233 | Employees Left: 237
- Attrition Rate: 16.12% | Average Age: 36.92 | Average Job Satisfaction: 2.73
- Average Monthly Income: $6,503 | Average Years at Company: 7.01

Charts included: Workforce by Department · Workforce by Gender · Attrition by Department · Employees by Age Group · Employees by Education · Business Travel Distribution · Workforce by Job Role



Dashboard 2 — HR Attrition Analysis

Purpose: Identify patterns associated with employee turnover across multiple dimensions.

Key Metrics:

- Employees Left: 237 | Attrition Rate: 16.12%
- Avg Years at Company (Left): 5.13 | Avg Age (Left): 33.61

Charts included: Attrition by Department · Job Role · Overtime · Marital Status · Education Field · Gender · Age Group · Income Band · Years at Company · Distance from Home



Dashboard 3 — Performance, Compensation & Satisfaction

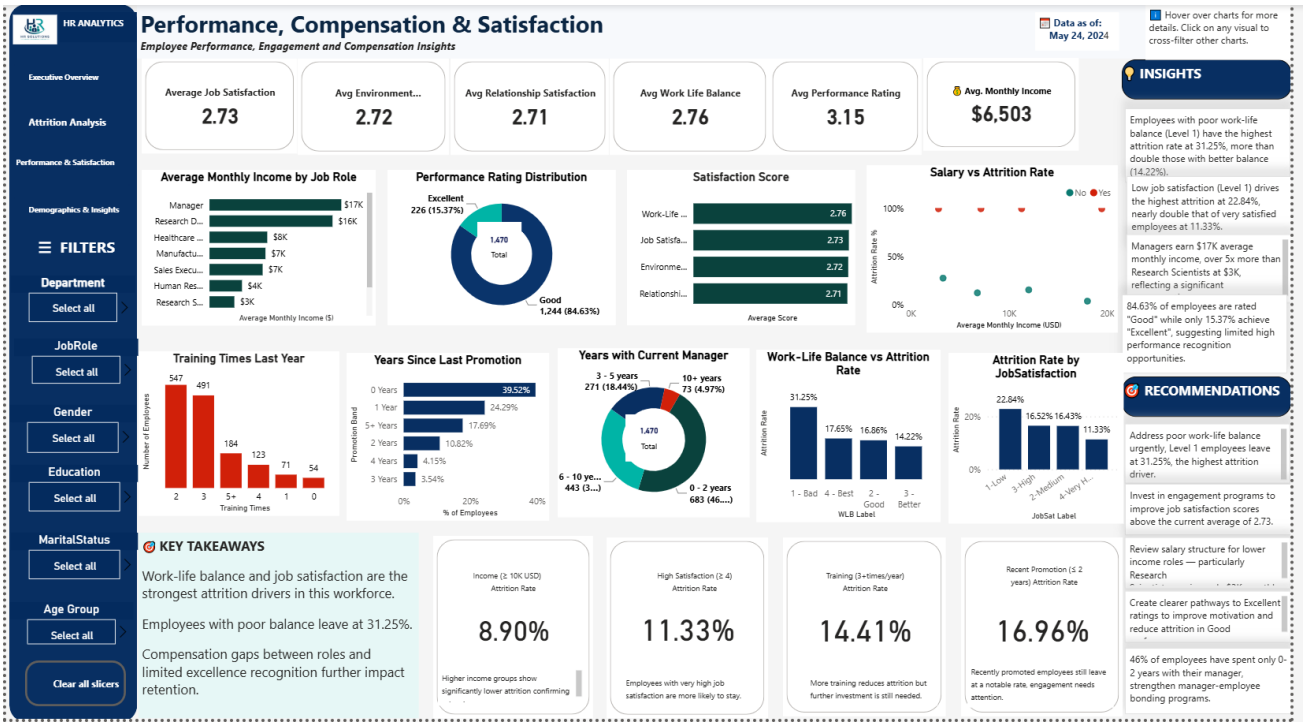
Purpose: Evaluate how performance, satisfaction, and compensation influence employee retention.

Key Metrics:

- Avg Job Satisfaction: 2.73 | Avg Environment Satisfaction: 2.72
- Avg Work-Life Balance: 2.76 | Avg Performance Rating: 3.15 | Avg Monthly Income: $6,503

Charts included: Income by Job Role · Performance Distribution · Satisfaction Scores · Salary vs Attrition · Training Times · Years Since Promotion · Years with Manager · WLB vs Attrition · JobSat vs Attrition

Key Takeaway Metrics: High Satisfaction Attrition (11.33%) · Recent Promotion Attrition (16.96%) · High Training Attrition (14.41%) · High Income Attrition (8.90%)

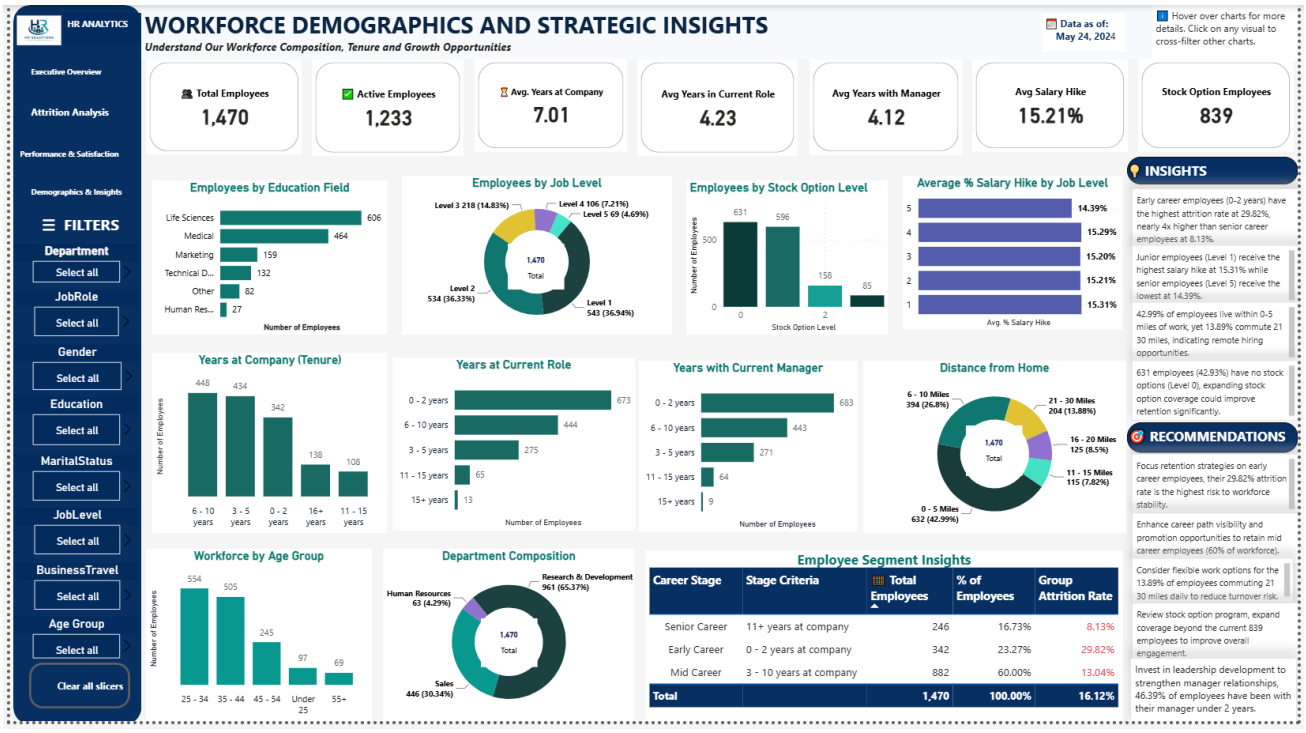


Dashboard 4 — Workforce Demographics & Strategic Insights

Purpose: Support workforce planning through employee segmentation and long-term demographic analysis.

Key Metrics:

- Avg Years in Current Role: 4.23 | Avg Years with Manager: 4.12
- Avg Salary Hike: 15.21% | Stock Option Employees: 839



Charts included: Education Field · Job Level · Stock Option Level · Salary Hike by Level · Tenure · Years in Role · Years with Manager · Distance from Home · Age Group · Department Composition · Employee Segment Insights Table

8. Business Questions Answered

The analysis addressed several important business questions:

Finding	Detail
Which departments have highest turnover?	Sales had the highest attrition at 20.63%, followed by Human Resources at 19.05%
Which job roles need retention focus?	Laboratory Technicians and Sales Executives showed the highest attrition rates
Does overtime increase attrition?	Yes — overtime employees had a significantly higher attrition rate at 28.57%
Does satisfaction influence retention?	Yes — employees with low satisfaction (Level 1) leave at 22.84% vs 11.33% for very satisfied employees
Are promotions linked to retention?	Employees promoted within 2 years still leave at 16.96% — engagement beyond promotion is needed
Which groups are at greatest risk?	Early career employees (0-2 years) have the highest attrition at 29.82%
How does income affect attrition?	Higher income employees (≥ $10K) have significantly lower attrition at 8.90%
What is the workforce composition?	R&D dominates at 65.37%, Life Sciences education leads at 41.22%, 25-34 age group is the largest at 42.18%

9. Key Findings

Attrition Findings

- Overall attrition rate is 16.12% with Sales showing the highest departmental rate at 20.63%
- Early career employees (0-2 years) have the most critical attrition rate at 29.82%
- Employees working overtime are significantly more likely to leave
- Longer commutes (31+ miles) correlate with higher attrition

Satisfaction & Performance Findings

- Average job satisfaction stands at only 2.73 out of 5 — indicating room for improvement
- Employees with poor work-life balance (Level 1) leave at 31.25% — the highest attrition driver
- Low job satisfaction (Level 1) drives attrition to 22.84% vs 11.33% for highly satisfied employees
- 84.63% of employees are rated "Good" while only 15.37% achieve "Excellent"

Compensation Findings

- Managers earn $17K monthly — over 5x more than Research Scientists at $3K
- Higher income employees (≥ $10K) show significantly lower attrition at 8.90%
- Salary hike percentages are relatively consistent across job levels (14.39% to 15.31%)

Demographic Findings

- 42.18% of the workforce is aged 25-34 — the largest and most strategically important group
- 65.37% of employees work in Research & Development
- 42.93% of employees (631) have no stock options — a potential retention lever
- 42.99% of employees live within 0-5 miles of work

10. Recommendations

#	Recommendation	Data Support
1	**Focus retention on Sales Department**	Sales attrition is 20.63% — the highest across all departments
2	**Prioritize early career employee engagement**	Employees with 0-2 years have the highest attrition at 29.82%
3	**Improve job satisfaction programs**	Average satisfaction is only 2.73 out of 5 across the workforce
4	**Address work-life balance urgently**	Poor WLB (Level 1) drives the highest attrition at 31.25%
5	**Review overtime policies**	Overtime employees show significantly higher attrition at 28.57%
6	**Expand stock option coverage**	631 employees (42.93%) currently have no stock options
7	**Develop millennial retention programs**	42.18% of the workforce is aged 25-34 — the largest at-risk group
8	**Review compensation for lower roles**	Research Scientists earn only $3K vs Manager average of $17K monthly
9	**Increase training frequency**	Employees with 3+ training sessions show lower attrition at 14.41%
10	**Monitor and track KPIs regularly**	Data-driven decisions require consistent monitoring of HR metrics

11. Business Value

This project demonstrates how HR analytics can help organizations create measurable impact:

Reduce Turnover Costs Identify and address attrition drivers before they escalate	**Improve Workforce Planning** Use demographic and tenure data for strategic HR decisions
Evidence-Based Decisions Replace assumptions with data-driven recommendations	**Monitor Performance** Track satisfaction, compensation and engagement metrics in real time

12. Conclusion

Data is most valuable when it informs action.

This HR Analytics project transforms 1,470 employee records into meaningful insights that help organizations better understand their workforce, identify retention challenges, and support strategic decision-making across four critical HR dimensions.

By combining interactive Power BI dashboards with business-focused analysis, the project moves beyond simple reporting to create measurable organizational value.

Good dashboards display information. Great dashboards answer questions, support decisions, and drive meaningful action.

Skills Demonstrated

Data Cleaning	**Data Modeling**	**DAX Development**
Data Visualization	**Dashboard Design**	**HR Analytics**
Business Intelligence	**KPI Development**	**Data Storytelling**
Insight Generation	**Executive Reporting**	**Recommendation Development**

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